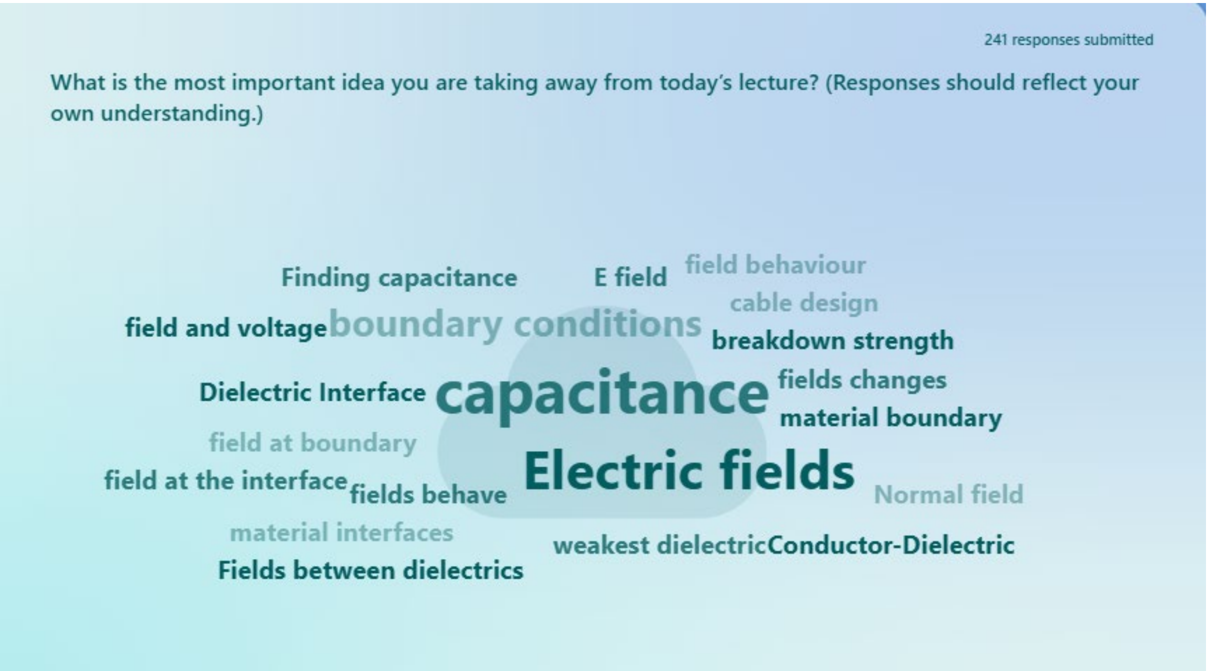
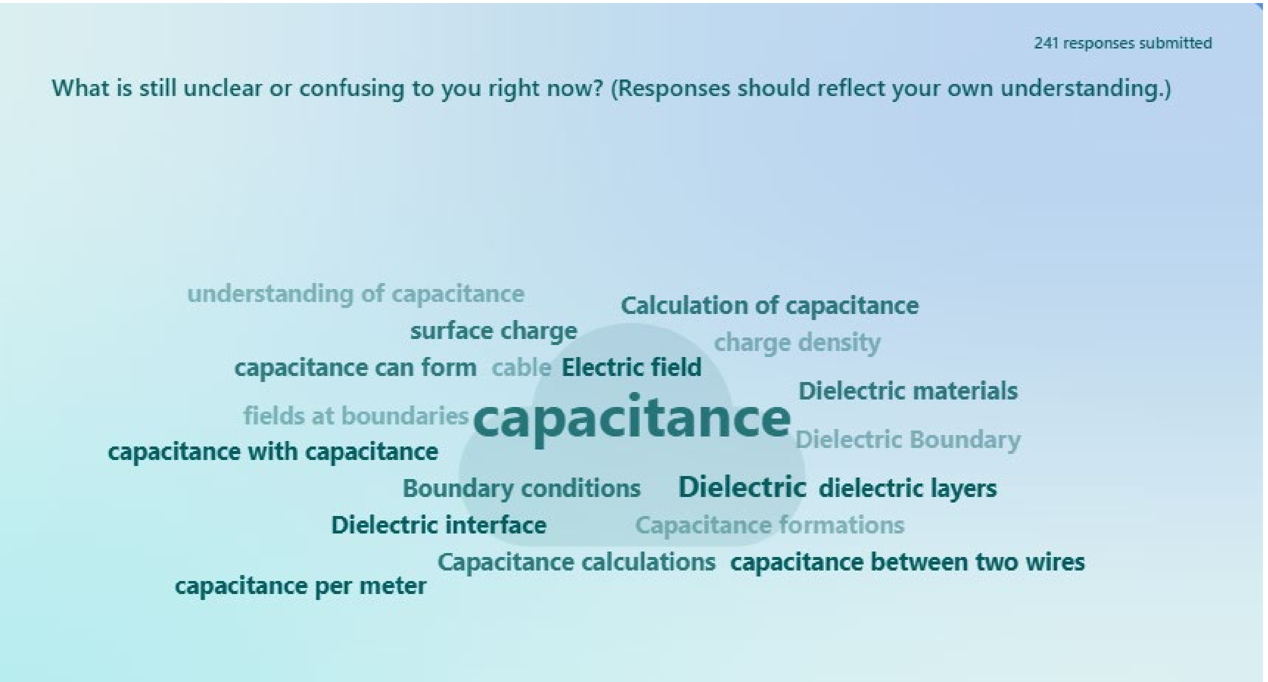
ELEC 3115

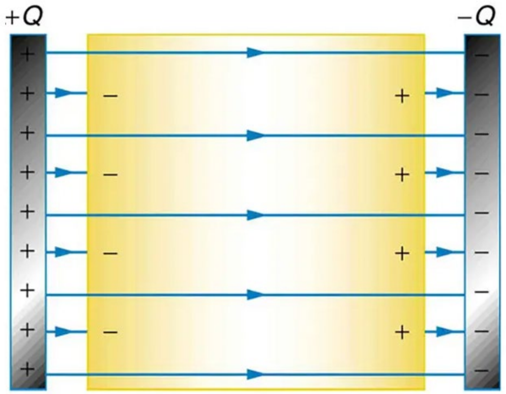
Lecture 4: Week 2 , Thursday

Your responses at the end of lecture 3:





Capacitance formation: How it relates to electric field (conceptually)



1. What does charge actually do?

When we place +Q and – Q on conductors → An electric field forms in the space between them. For example, in a parallel plate system,

$$|E| = \frac{Q}{\varepsilon S}$$

Key idea:
More charge → stronger electric field
Material permittivity ε controls field strength
Electric field is the physical effect of charge in space.

2. An electric field creates voltage → accumulation of electric field.

$$V = -\int_{2}^{1} \vec{E} \bullet d\vec{l} = \frac{Qd}{S\varepsilon}$$

(for the parallel plate example)

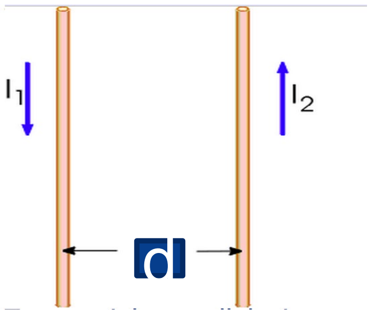
Stronger E→ larger V, larger separation d → larger V
Voltage is the consequence of the electric field.

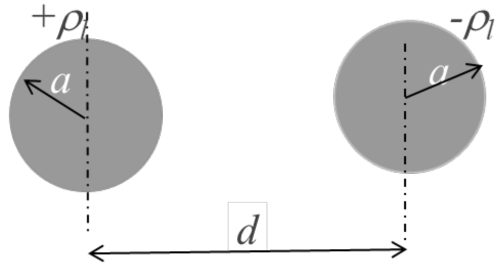
3. What is capacitance? $C = \dfrac{Q}{V}$

$$Q = CV$$

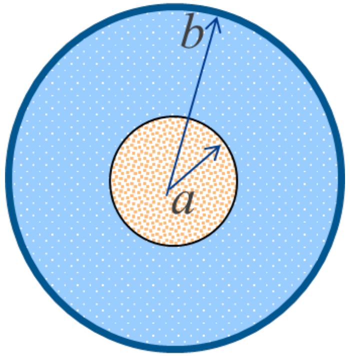
Capacitance tells us how much charge is required to create a given electric field (and voltage).
It is a property of how geometry and material shape the electric field: how easily they allow the electric field to form.

Whenever there are charges that produce *E* and hence *V*, we will have C

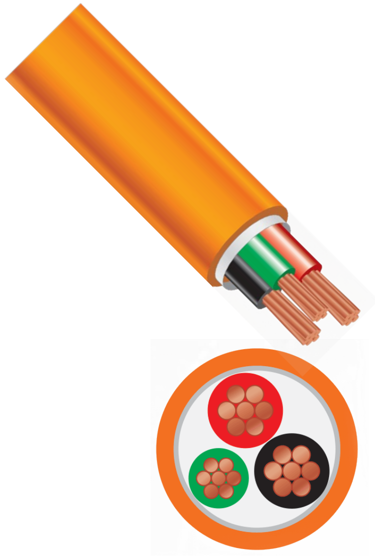




$$C = \frac{\rho_l}{V_{12}} = \frac{\pi\varepsilon_0}{\ln \frac{d-a}{a}} \text{ F/m}$$



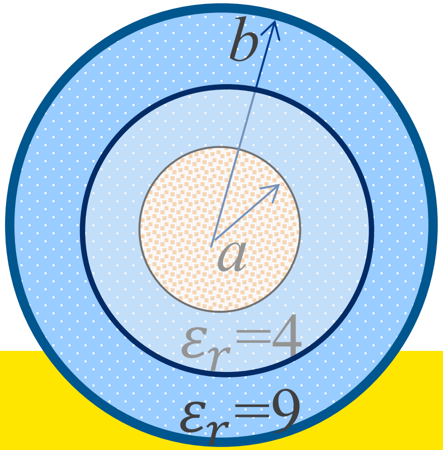
$$C = \frac{\rho_l}{V_{ab}} = \frac{2\pi\varepsilon}{ln\left(\frac{b}{a}\right)} F/m$$



Example 6

A long co-axial cable is made with a conductor of radius a = 2 mm and shield internal radius b = 5 mm. There are two layers of insulation between the conductor and the shield. The first insulation layer is 1mm thick and made of rubber (ε_r= 4). The second layer is plastic (ε_r=9) and 2 mm thick. Calculate the capacitance per unit length of the cable.

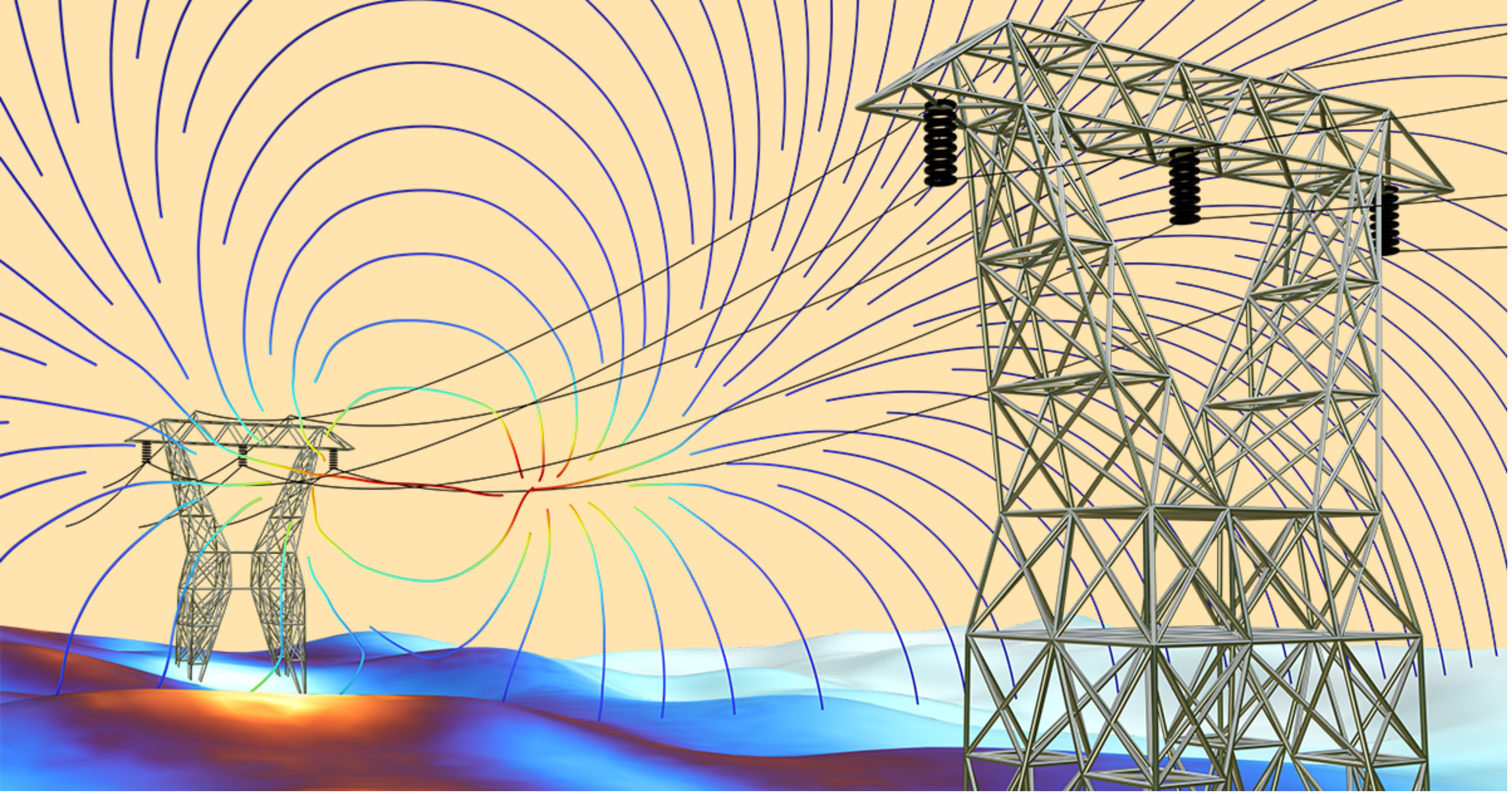
Ans: 352pF/m

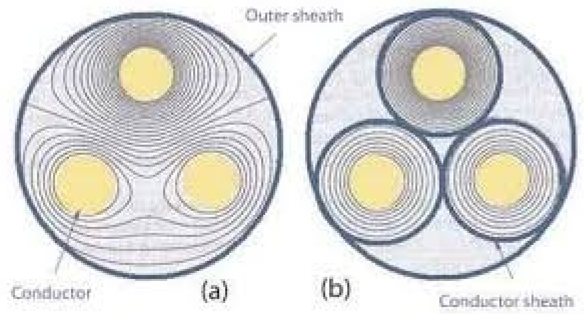


1. Field $\vec{E}_r = \vec{a}_r \dfrac{\rho_l}{2\pi\varepsilon r}$ V/m, $\varepsilon = \varepsilon_0\varepsilon_r, \varepsilon_0 = 8.854\times10^{-12}$ F/m

2. Potential $V_{ab} = -\int_b^{b-thickness} \vec{E}_{plastic} \bullet d\vec{r} - \int_{b-thickness}^{a} \vec{E}_{rubber} \bullet d\vec{r}$

2. capacitance $C = \dfrac{\rho_l}{V_{ab}}$





How to analyse the **capacitance network** of multiconductor systems such as multicore cables or multi-conductor transmission lines?

What is the effect of capacitance formation? Can we prevent unwanted capacitance coupling?

The capacitor stores electrostatic energy. Can this energy be used to do practical work by converting it to an electrostatic force? Can we calculate electrostatic force from energy directly?

Learning goals

By the end of today's lecture, you will be able to:

- Analyse and calculate capacitance formation in a multi-conductor system using the fundamental relation of $Q = CV$.

- Explain and assess techniques that prevent electrostatic interference in practical applications.

- Calculate electrostatic force from electrostatic energy using the concept of *virtual displacement*.

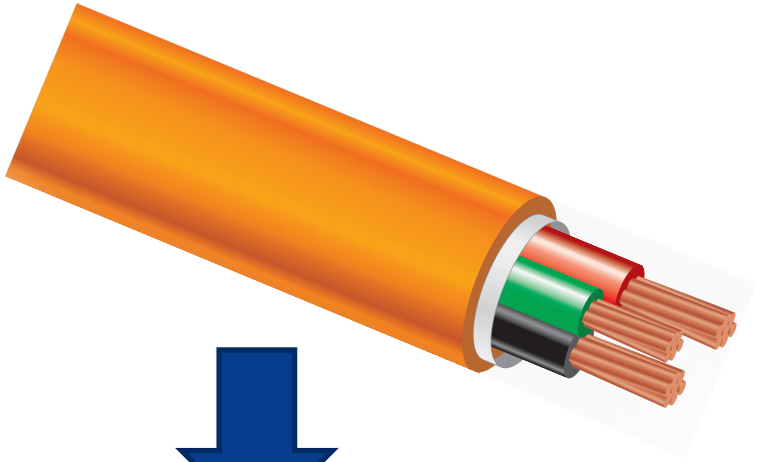
Capacitance in Multi-Conductor Systems

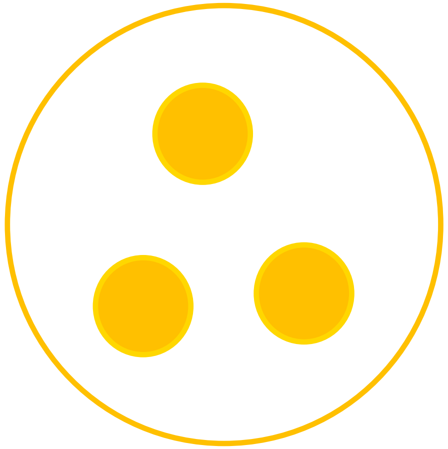
Real cables and PCBs have multiple conductors.
Each conductor interacts with all others.

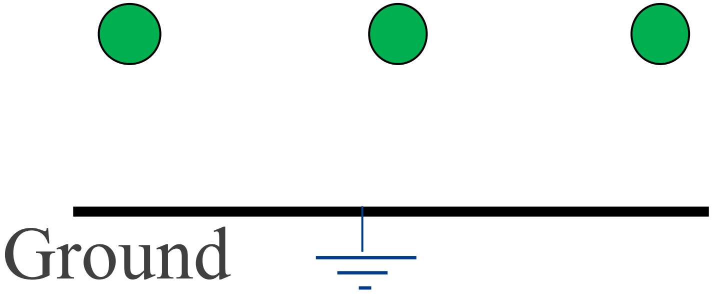
In this section, you will:

- Extend $Q = CV$ to multiple conductors
- Interpret capacitance matrices physically
- Understand self and mutual capacitance

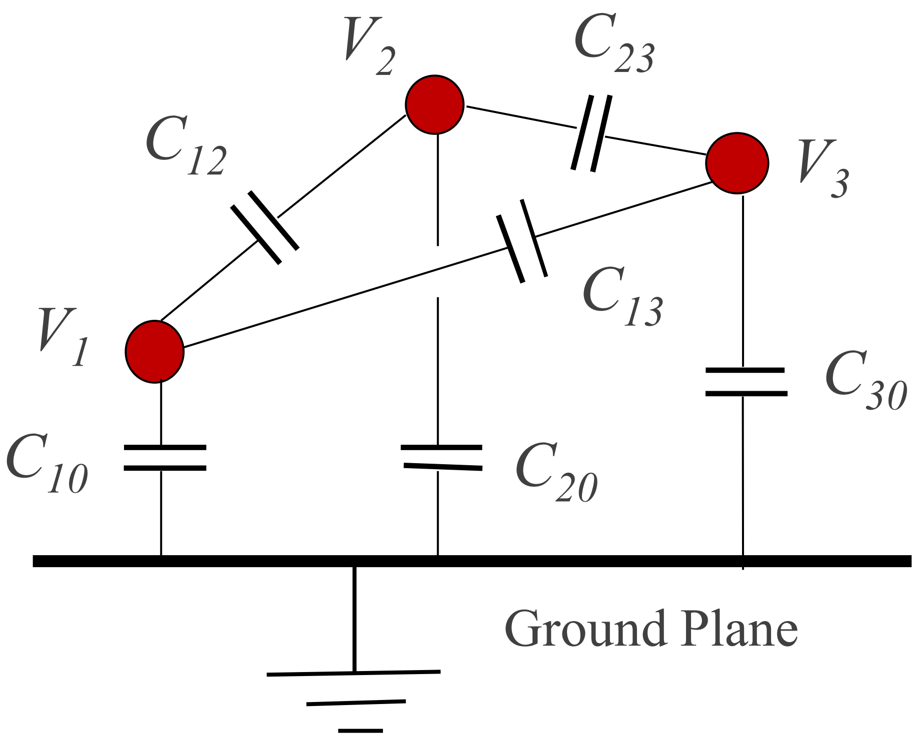
Case 4 :Capacitance of a multi-conductor system







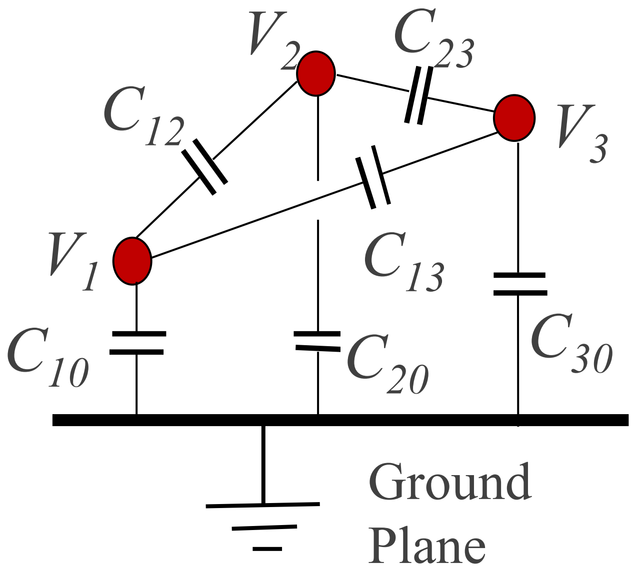
A 3-conductor system with a ground plane



- Consider a 3-conductor system above a ground plane
- V_1, V_2 and V_3 are the potentials of the three conductors.
- Q_1, Q_2 and Q_3 are the total charge of each conductor.

$$Q = CV \quad \Rightarrow \quad \begin{aligned} Q_1 &= C_{10}V_1 + C_{12}\left(V_1 - V_2\right) + C_{13}\left(V_1 - V_3\right) \\ Q_2 &= C_{20}V_2 + C_{12}\left(V_2 - V_1\right) + C_{23}\left(V_2 - V_3\right) \\ Q_3 &= C_{30}V_3 + C_{13}\left(V_3 - V_1\right) + C_{32}\left(V_3 - V_2\right) \end{aligned}$$

$$Q_1 = C_{10}V_1 + C_{12}(V_1 - V_2) + C_{13}(V_1 - V_3)$$
$$Q_2 = C_{20}V_2 + C_{12}(V_2 - V_1) + C_{23}(V_2 - V_3)$$
$$Q_3 = C_{30}V_3 + C_{13}(V_3 - V_1) + C_{32}(V_3 - V_2)$$



By rearranging,

$$Q_1 = (C_{10} + C_{12} + C_{13})V_1 - C_{12}V_2 - C_{13}V_3$$
$$Q_2 = -C_{12}V_1 + (C_{20} + C_{12} + C_{23})V_2 - C_{23}V_3$$
$$Q_3 = -C_{13}V_1 - C_{23}V_2 + (C_{30} + C_{13} + C_{23})V_3$$

In matrix form,

$$\begin{bmatrix} Q_1 \\ Q_2 \\ Q_3 \end{bmatrix} = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{pmatrix} (C_{10} + C_{12} + C_{13}) & -C_{12} & -C_{13} \\ -C_{12} & (C_{20} + C_{12} + C_{23}) & -C_{23} \\ -C_{13} & -C_{23} & (C_{30} + C_{13} + C_{23}) \end{pmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

This analysis can be generalized for N number of conductors,

$$\begin{bmatrix} Q_1 \\ Q_2 \\ . \\ . \\ Q_N \end{bmatrix} = \begin{bmatrix} c_{11} \; c_{12} \; \ldots\ldots\ldots\ldots c_{1N} \\ c_{21} \; c_{22} \; \ldots\ldots\ldots\ldots c_{2N} \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ c_{N1} \; c_{N2} \ldots\ldots\ldots\ldots c_{NN} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ . \\ . \\ V_N \end{bmatrix}$$

$$\boxed{\begin{array}{l} c_{11} = \left(C_{10} + C_{12} + \ldots + C_{1N}\right), \\ . \\ . \\ c_{NN} = \left(C_{N0} + C_{N2} + \ldots + C_{NN}\right) \end{array}}$$

$$\boxed{\begin{array}{l} c_{12} = c_{21} = -C_{12}, \\ . \\ . \\ c_{1N} = c_{N1} = -C_{1N}, \end{array}}$$

For a N conductors we get a N by N capacitance matrix.

⚠ Exam Signposting — Multi-Conductor Systems

In exams, you may be asked to:

✔ Construct the capacitance matrix from given mutual capacitances

✔ Interpret diagonal and off-diagonal terms

✔ Compute unknown voltages or charges

You are expected to:

- Understand physical meaning of coefficients
- Not memorise matrix form blindly

Coefficient of capacitance: diagonal elements of the capacitance matrix

$$c_{ii} = \frac{Q_i}{V_i}, i = 1, 2, 3...$$ (Keeping all other conductors at ground.)

It is the total capacitance in a conductor when all other surrounding conductors and ground are at zero potential.

e.g. $c_{11} = (C_{10} + C_{12} + C_{13}), \quad c_{22} = (C_{20} + C_{12} + C_{23}), \quad c_{33} = (C_{30} + C_{13} + C_{23})$

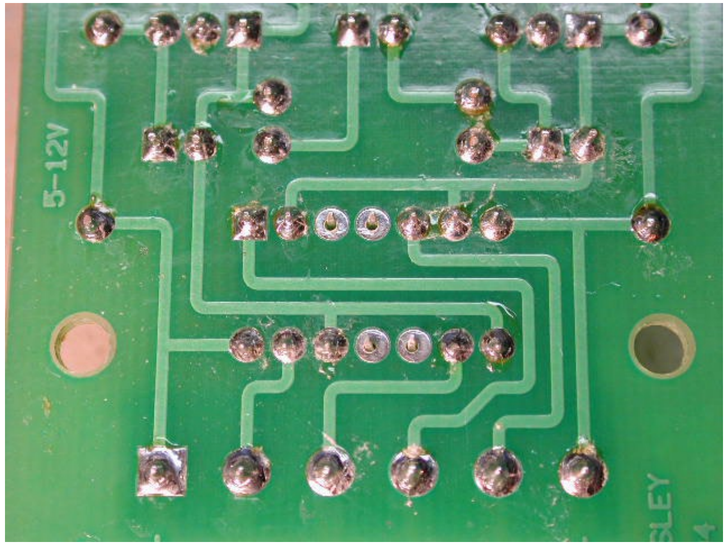
The element c_{ij} $(i \neq$ j)of the capacitance matrix equals to the negative of the <u>mutual capacitances</u> between two adjacent conductors *i* and *j*.

e.g. $c_{12} = c_{21} = -C_{12}, \quad c_{23} = c_{32} = -C_{23}, \quad c_{13} = c_{31} = -C_{13},$

Unwanted Capacitance & Electrostatic Shielding

Capacitance can store energy — but it can also cause interference.

In this section, you will:
•Analyse capacitive coupling
•Interpret capacitance matrix terms physically
•Understand how grounding eliminates coupling



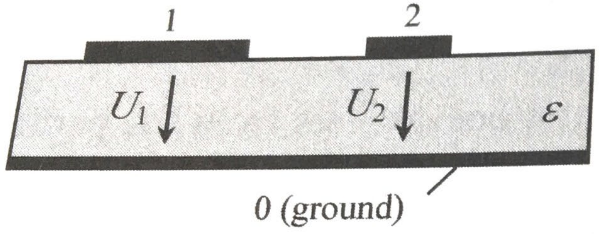
Example 7

A cross-sectional view of a printed circuit board (PCB) consisting of two conducting tracks on the surface of a dielectric board above a reference ground plane is shown in Figure (a). The capacitive coupling of the system is shown in Fig.(b). The values of the mutual capacitances are found experimentally as shown in fig. (b).

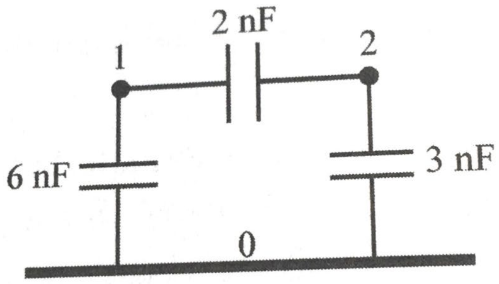
(i) Find the capacitance matrix $[c]$.

(ii) If track 1 is at potential $U_1 = 10$ V and the track 2 is idle (i.e. $Q_2 = 0$). Find the resulting values of Q_1 and U_2.

(iii) If track 2 potential $U_2 = 0$ and track 1 is isolated, will there be any charge or voltage in the system?



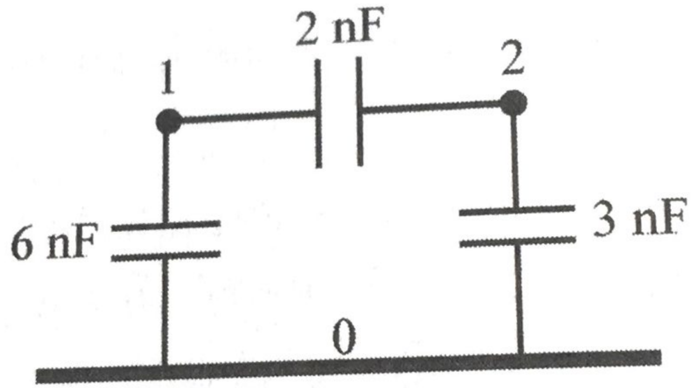
(a)



(b)

Solution

(i) Find the capacitance matrix $[c]$.



How many rows and columns the capacitance matrix will have?

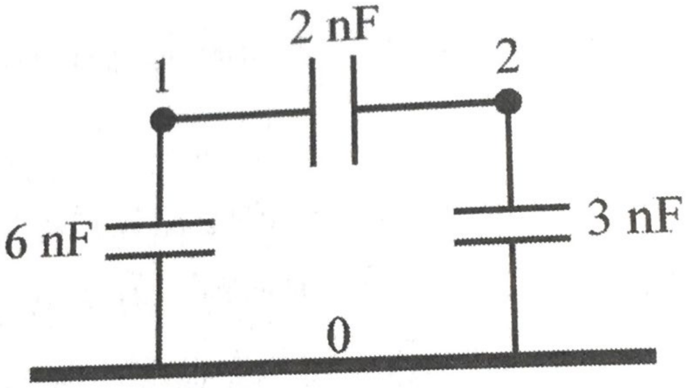
$$[c] = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix} \qquad \therefore [c] = \begin{bmatrix} 8 & -2 \\ -2 & 5 \end{bmatrix} \mathrm{nF}$$

$$c_{11} = C_{10} + C_{12} = (6+2)\ nF$$
$$c_{22} = C_{20} + C_{21} = (3+2)\,nF$$

$$c_{12} = c_{21} = -C_{12} = -2\ \mathrm{nF}$$

UNSW SYDNEY

(ii) If track 1 is at potential U_1 = 10 V and the track 2 is idle (i.e. Q_2 =0). Find the resulting values of Q_1 and U_2.



$$\begin{bmatrix} Q_1 \\ Q_2 \end{bmatrix} = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix} \begin{bmatrix} U_1 \\ U_2 \end{bmatrix} \Rightarrow \begin{bmatrix} Q_1 \\ 0 \end{bmatrix} = \begin{bmatrix} 8\times 10^{-9} & -2\times 10^{-9} \\ -2\times 10^{-9} & 5\times 10^{-9} \end{bmatrix} \begin{bmatrix} 10 \\ U_2 \end{bmatrix}$$

$$Q_1 = c_{11}U_1 + c_{12}U_2$$
$$Q_2 = c_{21}U_1 + c_{22}U_2$$

U_1 = 10 V and Q_2 =0, Find Q_1 and U_2

Ans: Q_1 = 72nC and U_2= 4V

(iii) If track 2 is at zero potential (it may be connected to the ground) and track 1 is isolated, will there be any charge or voltage in the system?

The isolated track 1 has charge Q_1 = 72 nC (Charges remain unless grounded and discharged). Track 2 is grounded i.e. U_2=0

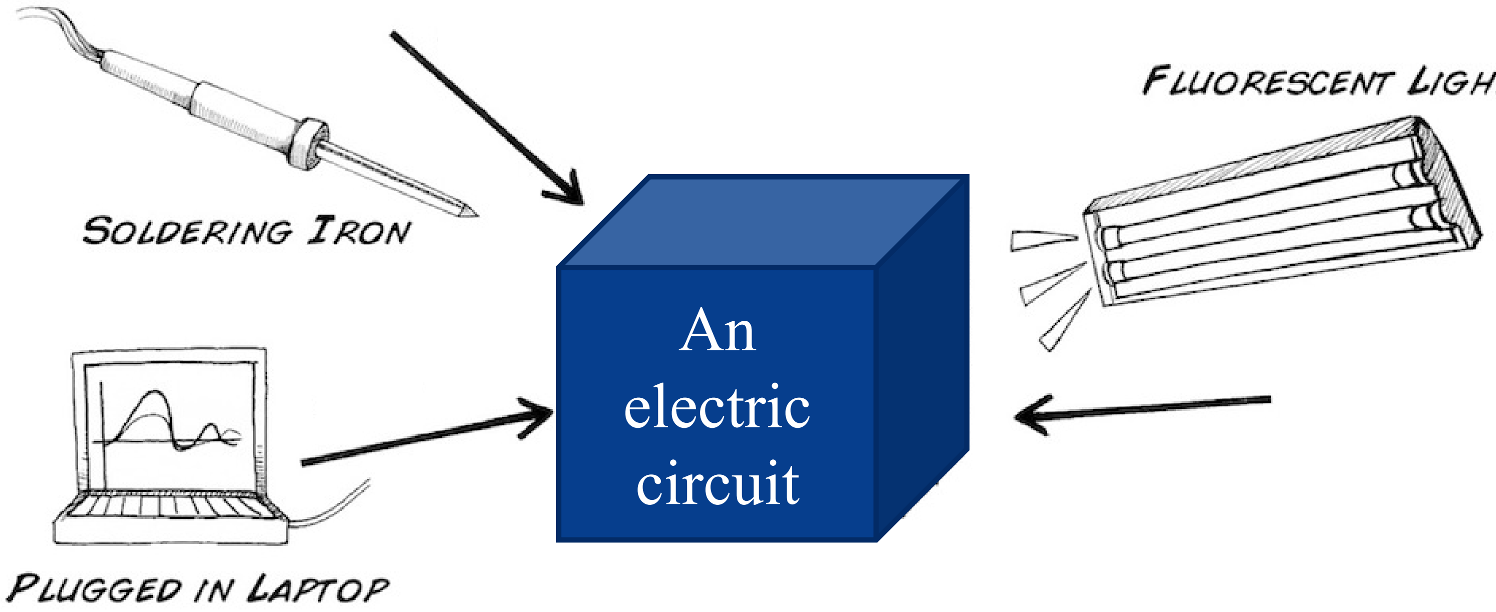
$$\begin{bmatrix} Q_1 \\ Q_2 \end{bmatrix} = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix} \begin{bmatrix} U_1 \\ U_2 \end{bmatrix} \Rightarrow \begin{aligned} Q_1 &= c_{11}U_1 + c_{12}U_2 \\ Q_2 &= c_{21}U_1 + c_{22}U_2 \end{aligned}$$

$$[c] = \begin{bmatrix} 8 & -2 \\ -2 & 5 \end{bmatrix} \text{nF}$$

Find Q_2 and U_1

Ans: Q_2 = -18nC and U_1 = 9V

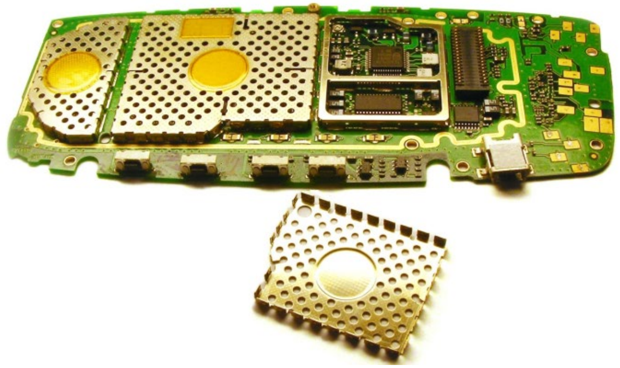
Unwanted capacitance formation:



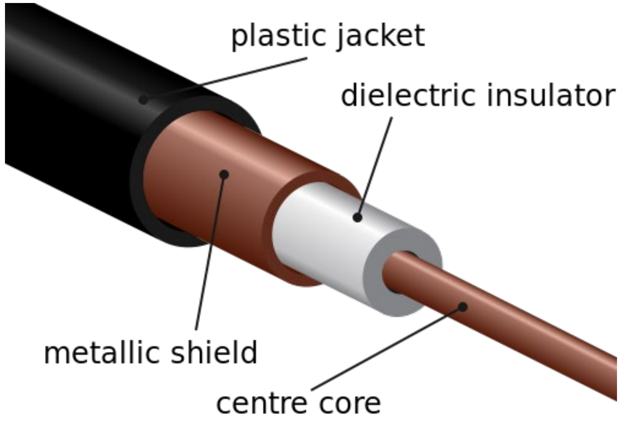
Can we prevent unwanted capacitance formation in a system?

Electrostatic shielding

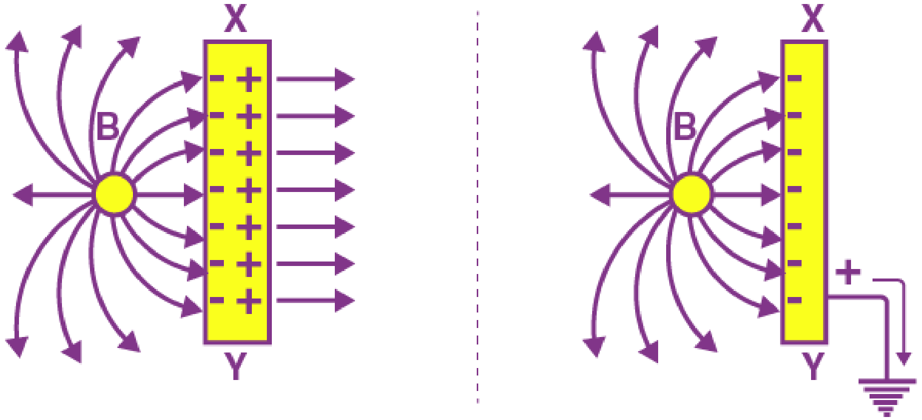
It is a technique to reduce the effect of external electrical field (i.e., capacitive coupling between conducting bodies) in a space by blocking the field with conductive barriers (i.e., **Faraday's cage**). It is commonly used to protect electronic equipment from lightning and other external electrostatic discharges and interference.



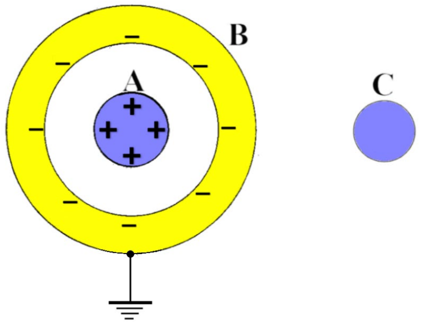
Shielding cage in a mobile phone



Shield in a coaxial cable



Shielding when the screen is grounded.



The grounded conductor B shields the electric field of A, hence no capacitance coupling between A and C.

⚠ Exam Signposting — Shielding & Coupling

You may be required to:

✔ Explain how grounding eliminates capacitive coupling

✔ Predict behaviour when a conductor is isolated vs grounded

✔ Interpret coupling using matrix elements

Pause & Reflect (Before we take a short break…)

✏ **Take 1 minute to jot down:**

1. **One key idea** from this part of the lecture
2. **One question** that is still unclear
3. **One connection** to something you already know
   (another course, lab, real-world system, intuition)

💡 *This is for you — no submission.*

Energy Stored in the Electric Field

Capacitors store energy in space, not in the plates.

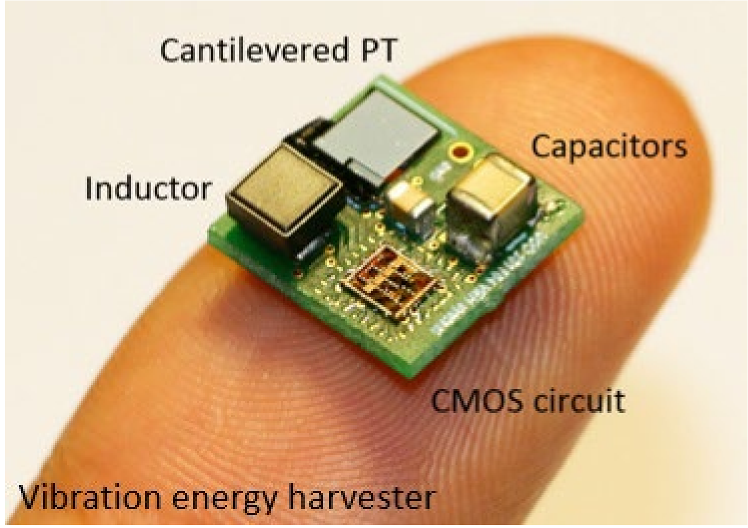
In this section, you will:

- Derive electrostatic energy expressions
- Understand energy density
- Connect energy to electric field strength

What are they doing here?

Can we use electrostatic force to do useful work?

Examples: Micro-machine or MEMS: (Micro-Electro-Mechanical Systems)
MEMS is very small device in which mechanical and electrical functions are combined on a chip. It may have capacitive actuators, sensors and transducers.



Electrostatic Energy and Force

Capacitor stores energy in the electric field.

$$W_e = \frac{1}{2} C V^2 \quad \text{Joules}$$

and because $C = \dfrac{Q}{V}$

$$W_e = \frac{1}{2} Q V \quad \text{J}$$

$$\because Q = \int_{vol} (\rho dv)$$

$$W_e = \frac{1}{2} V \int_{vol} (\rho dv) \quad [\text{J}]$$

$$W_e = \frac{1}{2} V \int_{vol} (\rho dv) \quad \text{[J]}$$

We know, $\vec{\nabla} \bullet \vec{D} = \rho$ and $-\vec{\nabla} V = \vec{E}$

It can be shown that , $W_e = \frac{1}{2} \int_{vol} \left(\vec{D} \bullet \vec{E} \right) dv \quad \text{[J]}$

For full derivation refer to page 137 of your textbook by D. K. Cheng.

$$W_e = \left[\frac{1}{2} \vec{D} \bullet \vec{E} \right] [Volume] \Rightarrow \boxed{w_e = \frac{1}{2} \vec{D} \bullet \vec{E} = \frac{1}{2} \varepsilon E^2 = \frac{1}{2} \frac{D^2}{\varepsilon} \quad [\text{J/m}^3], \ \vec{D} = \varepsilon \vec{E}}$$

Energy density w_e (i.e. energy per unit volume) is determined from **E** and **D**.

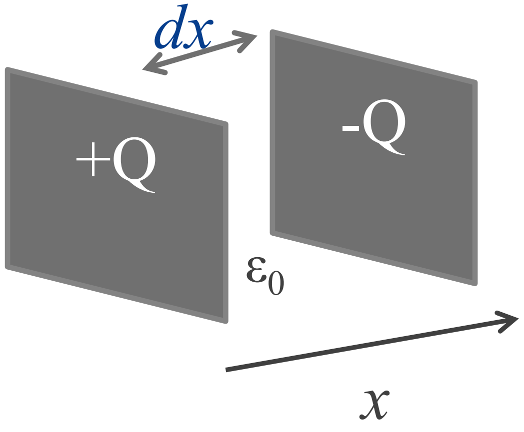
Electrostatic Force from Energy

Electric fields can perform mechanical work.

In this section, you will:

- Apply conservation of energy
- Distinguish constant-Q and constant-V systems
- Use virtual displacement to compute force

Electrostatic Force from Energy



Plates attract each other – why?

How do we calculate this attractive force using electric field theory?

We can calculate force from how energy changes.

In parallel plates capacitor, $W_e = \frac{1}{2} C V^2$ Joules $\quad C = \frac{\varepsilon S}{d}$

If the plates moves due the force, energy will increase or decrease depending on the direction of movement (i.e., the force). This means force is the gradient of energy.

$$\vec{F}_e = \pm \vec{\nabla} W_e$$

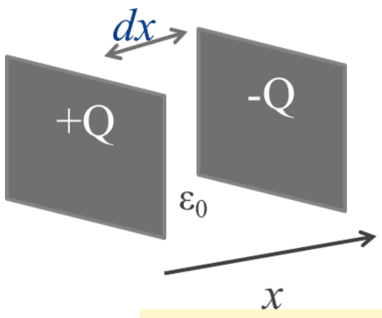
To find the force, the plates do not physically need to move. We could consider a virtual (i.e., hypothetical) tiny movement. -Calculation of force by the virtual displacement method.

The force found by this method considers that systems tend to move in the direction of reduced energy. Hence, in the parallel plate, the force found from virtual displacement prevents *d* from decreasing (i.e., the equivalent repulsive force that must be applied to prevent the plate from moving closer).

How to apply $\vec{F}_e = \pm\vec{\nabla} W_e$

Stored energy $W_{e,}$ is a function of both Q and V.

$$W_e = \frac{1}{2} QV \quad \text{J}$$



Gradient calculation becomes simplified if you consider either Q or V constant in a system.

$$C = \frac{\varepsilon S}{d = x}$$

Let's consider parallel plates as our example case,

Constant Q System

plates are not connected to a voltage source

$$W_e = \frac{1}{2} QV, C = \frac{Q}{V}, \therefore W_e = \frac{1}{2} Q\left(\frac{Q}{C}\right) = \frac{1}{2}\frac{Q^2}{C}$$

If the plates move closer, C increases $\rightarrow W_e$ decreases

Force is the negative gradient of W_e. $\vec{F}_e = -\vec{\nabla} W_e$

Parallel plate example:

$$\vec{F}_x = -\frac{dW_e}{dx} = -\frac{1}{2} Q^2 \frac{d}{dx}\left(\frac{x}{\varepsilon S}\right) = -\frac{1}{2} QE$$

Constant V System

plates are connected to a voltage source (i.e., V is constant)

$$W_e = W_e = \frac{1}{2} QV, C = \frac{Q}{V}, \therefore W_e = \frac{1}{2} CV^2$$

If the plates move closer, C increases $\rightarrow W_e$ increases

We increase because the external source V replenishes energy.

Force is the positive gradient of W_e. $\vec{F}_e = \vec{\nabla} W_e$

$$\vec{F}_x = \frac{dW_e}{dx} = \frac{1}{2} V^2 \frac{d}{dx}\left(\frac{\varepsilon S}{x}\right) = -\frac{1}{2} QE$$

Both systems give the same force that prevents a change in energy

UNSW SYDNEY

Homework

⚠ Exam Signposting — Energy & Virtual Displacement

You may be asked to:

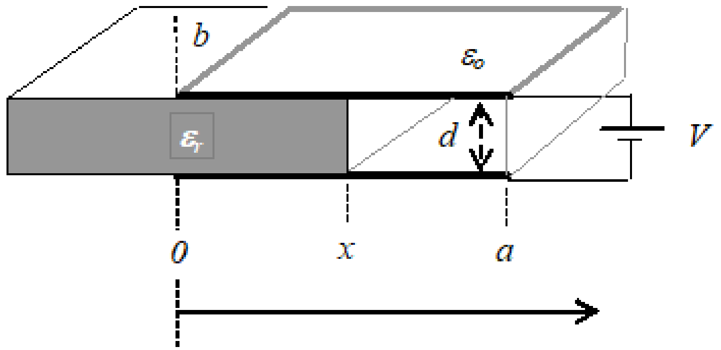
✔ Derive force from energy expression

✔ Distinguish constant-Q and constant-V cases

✔ Interpret the sign of the force

Common mistake:

✖ Mixing up sign convention for constant-Q vs constant-V systems



Consider the parallel plate capacitor in which the dielectric material between the plates is displaced from the vertical line at $x = 0$, as indicated in the figure. The width of the plate is b, and the distance between the two plates is d.

(i) Show that capacitance between the plates $C = \frac{\varepsilon_0 b}{d}\{a + (\varepsilon_r - 1)x\}$

(ii) Show that the force acting on the dielectric when a fixed voltage V is applied to the plates is $\vec{F}_x = \frac{V^2}{2}\frac{\varepsilon_0 b(\varepsilon_r - 1)}{d}\,\vec{a}_x$

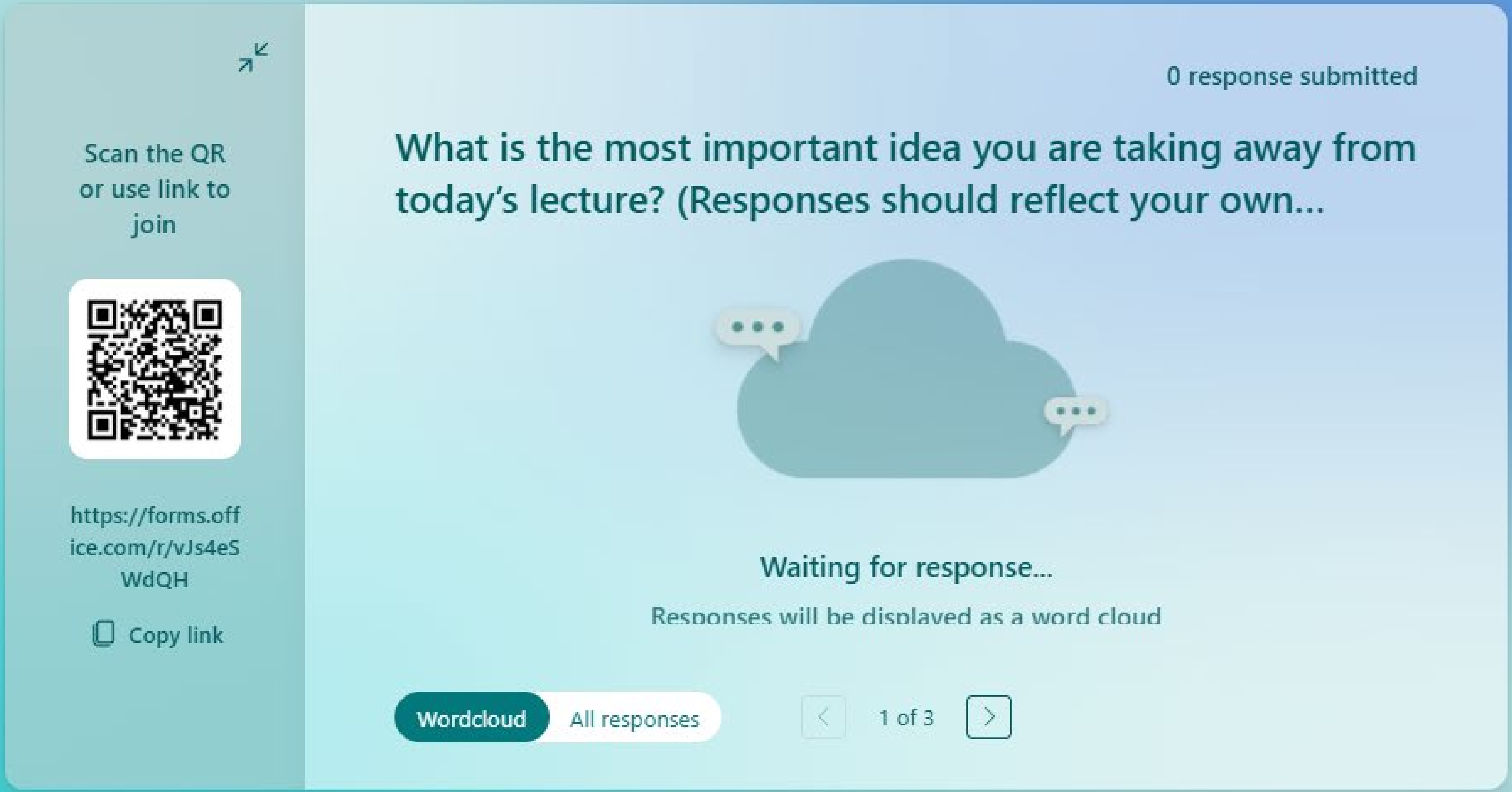
(iii) Calculate the magnitude of the force for the following numerical values:

$\varepsilon_r = 5$, $a = 12$ mm, $b = 18$ mm, $d = 2$ mm, $V = 10$ V.

Ans: 15.9 nN

Before You Go… (1 minute reflection)

This reflection will count toward optional bonus marks.



Key ideas to take away from today

1 Capacitance is a Field Property

Capacitance depends on:

•Geometry, Permittivity, and Electric field distribution

It measures how much charge is needed to produce a given voltage

2 Multi-Conductor Systems Generalise $Q = CV$

For multiple conductors:

$$\mathbf{Q} = [\mathbf{C}]\mathbf{V}$$

•Diagonal terms → self-capacitance

•Off-diagonal terms → mutual coupling

Capacitance matrix describes interaction between conductors.

3 Capacitance Can Be Useful — or Unwanted

•Useful: energy storage

•Unwanted: capacitive coupling

- Grounded shielding eliminates unwanted coupling.

4 Energy Is Stored in the Electric Field

$$W = \frac{1}{2}\int \varepsilon E^2 \, dv$$

Energy is stored in space, not in the conductors.

5 Electric Fields Can Produce Force

Electrostatic force is obtained from energy:

•Constant Q → $F = -\nabla W$

•Constant V → $F = +\nabla W$

Sign depends on energy flow.

Review and further practice (Recommended)

Practice Examples 3-17, 3-18, 3-19 from the textbook by D.K Cheng

Try out Problems – P3-29, P3-30, P3-33, P3-35,

Challenge problems –P3-36, P3-40, P3-44, P3-46, P3-47, P3-48